Final Report: DATASCI 192A/B

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Introduction

The San Francisco Giants are one of the most storied franchises in Major League Baseball (MLB) with 8 World Series Titles, 23 National League Pennants, more wins than any other team in the MLB, and over 140 years of history. In order to further their legacy, the Giants go to great lengths to be as competitive as possible throughout the season. Players follow strict workout regimens, coaches watch hours of film on opposing bullpens, and front offices work around the clock to secure the best players. However, something no amount of discipline or planning can control is game day weather. Our project identifies data-driven trends in the variability of daily weather to help our partner, the Giants, better prepare for game-day success.

Baseball Savant aggregates stadium trends into the “park factor” statistic, which tracks the rate of runs at a given stadium versus a theoretical neutral ground. However, this statistic is not a complete picture of how each park influences play, as environmental factors can manifest in different ways depending on the stadium. While park factor is insightful, this static metric tells us more about physical stadium characteristics than how to prepare for the day-to-day differences in conditions that really matter when your team takes the field. There are general phenomena that apply universally, like low pressure helping the ball carry farther, but the same weather event can have very different outcomes across stadiums. For example, an April night game at Dodger Stadium doesn’t have drastically different conditions than a day game, but 400 miles north at Oracle Park, spring night games often come with heavy fog rolling off the bay and powerful winds that severely change the way the ball moves. Today, the Giants do not have the right information to fit their game day strategy to expected conditions because existing statistics emphasize annual trends over the daily reality of the stadium.

In order to address this knowledge gap, we built MLB ParkCast, an interactive dashboard that will help the Giants get ready for game day through comprehensive stadium analysis. In addition to providing detailed information on the physical characteristics of every MLB stadium, like elevation, outfield shape and size, fence heights, and unique attributes (ex. Green Monster) that influence long-run trends of park play, this tool also analyzes the day-to-day impact of weather and game time to help your team strategize effectively. To use MLB ParkCast, the Giants input the stadium, game time, expected weather (temperature, pressure, wind conditions, precipitation), and date a few days before their game, and the dashboard will predict expected game-day park factor based on historical game and weather data.

Before the Giants take the field, they want to know how a heat wave in Arlington, snow at Fenway, or rain at Yankee stadium is going to impact the quality of play, and MLB ParkCast helps close this information gap through targeted analysis of stadium characteristics and game day conditions. This dashboard goes beyond static annual park factors by using predictive models to estimate strikeouts and runs on a specific day, and calculate numeric deltas for these stats compared to stadium and league averages. This information empowers coaches to make strategic decisions about which pitchers to play, how many relievers to prepare, how deep to play outfielders, who to put in batting line ups, and whether batters should focus on contact or home runs. We can't control the weather, but MLB ParkCast can help the Giants align their strategy to expected conditions to dominate on the field.

Literature Review

Prior research examines how stadium characteristics influence offensive production. Petriello (2021) introduces how park factors can be tracked on Baseball Savant and explores examples of how park characteristics can dramatically affect play. Park factor measures how much more or less offense is produced in a given park compared to a league-average stadium, where the league average is set to 100. For example, a park factor of 105 indicates that run scoring is 5% higher than average. Because stadiums vary in dimensions, altitude, and air density, park factors help standardize comparisons across parks. However, the metric does not capture player-specific differences, as players may respond differently depending on hitting style or handedness. Funf's (2025) central argument is that each stadium's physical and environmental characteristics change how the ball travels and how the players react. As a result, the same hit or pitch can produce different outcomes depending on the ballpark.

After solidifying our understanding of the park factor metric, we researched historical baseball data trends and weather outcomes we suspected would be influential, such as the relationship between temperature and hitting. The 2013 paper "The Impact of Temperature on Major League Baseball" by Koch & Panorska explores how temperature affects various baseball statistics, such as runs scored, batting average, slugging, home runs, and walks using a dataset of over 22,000 MLB games between 2000-2011. Major League Baseball is played across the United States (and Toronto, Canada), from April to October, so teams experience Spring, Summer, and Fall conditions throughout the country. Snowy fields in Boston or heat waves in Arlington impact not only the balls in play, but also the players on the field as temperature changes reaction time, air density, and player fatigue. Koch and Panorska found that home runs and runs scored increase significantly in warm weather compared to cold weather, with the effect more pronounced in the American League (NL did not have designated hitters yet). This paper is impactful for our project because it shares insight on how temperature impacts play, reminds us to consider how weather impacts both the ball and the player when thinking about park factors, and provides statistical methods to analyze how weather impacts play, such as Wilcoxon rank sum tests.

Many coastal MLB stadiums, such as Oracle Park and Angel Stadium are heavily impacted by the local marine layer. In order to better understand the effects of aquatic environments on baseball statistics, we referred to Kagan and Mitchell's research on "The effect of the Marine Layer on Fly Balls." The marine layer is moisty, humid, foggy air that blows in off the Pacific ocean – especially during summer nights – and is often accused of turning home runs into fly outs. While humid air is lighter, cold air is denser, so the on-paper effects of the marine layer are ambiguous. However, Kagan and Mitchell's regression model showed that the marine layer effect accounts for over 90% of the variance in fly ball distance with an average effect of around 6 feet (page 6). This provides convincing evidence that the marine layer is influential in turning home runs into fly balls and reducing carry at West Coast parks.

Wind also plays a key role in game-day conditions. Weather Applied Metrics in Major League Baseball by Nunnally (2023) details the statistics the MLB uses to analyze how wind impacts play. Nunnally explains how stadium weather data is different from what we see on our phone's weather app because the large structures surrounding the field distort wind patterns from what the flags flying atop the stadium indicate. The complex relationship of weather, stadium shape, and physics make wind one of the hardest park factor components to characterize. This article highlights Weather Applied Metrics (WAM) that are using computational fluid dynamic models to analyze wind on field and even make forecasts about how generic home run trajectories will be affected by weather on a given day. Nunnally's emphasis on the difference between ambient wind and the wind in the stadium impacting ball trajectory helped shape our team's wind analysis decisions. We projected wind along the vectors from home plate to center, left center, and right center field to better understand how wind would influence a batted ball's flight path.

While weather is a key determinant for gameday conditions, today, eight out of thirty MLB teams play in some form of covered stadium with a climate controlled environment. In order to better understand the "Dome Effect," we referenced Hunter's 2014 analysis of the Rogers Centre, home of the Toronto Blue Jays. In 2013, 9 out of 50 of the longest home runs were hit at the Rogers Centre, and of the 50 longest home runs at the Rogers Centre in the same season, 34 occurred when the roof was closed. On average, the Rogers Centre sees 2.9 home runs per game with the roof open and 2.4 home runs per game with the roof closed (out of 41 games played with the roof closed and 39 with the roof open), which is not substantially different. While it is notable that closed-roof home runs tend to be longer, this Rogers Centre analysis does not provide convincing evidence that domes substantially impact the likelihood of home runs.

Weather is clearly an influential factor in the game of baseball, however it is also an incredibly volatile independent variable. We conducted considerable research to better understand how to isolate the effects of weather on play and effectively predict daily park factor given game-time conditions. In 2025, Patterson published an article titled Modeling the Impact of the Atmosphere in Baseball: The Home Run Ball that explored how weather shapes baseball performance. Patterson looked how physics based modeling of temperature, pressure, humidity, and altitude would impact a simulated home run-like fly ball. He found that the same home run ball hit in 55℉ weather travels approximately 10 ft farther in 80℉, which corresponds to a 3-4ft gain for 10℉ increase in temperature on average. Further, lower air density (which reflects lower pressure and higher altitude) can cause a 2ft increase in carry for every 0.3 inHg decrease in pressure, which compounds to 30 extra feet at 1 mile of elevation compared to sea level. Patterson's humidity analysis was especially impactful for our project because it revealed that humid conditions actually create less drag because water vapor molecules are lighter than dry air molecules. However, while humidity can slightly increase carry, the ball also becomes more elastic and gains mass as it absorbs water vapor, which is an offsetting effect that makes humidity effects hard to isolate. While our model is more concerned with how ambient weather affects overall stadium conditions for any hitting, pitching, or fielding performance than a specific fly ball, Patterson's research helped us establish expected directionally for weather effects and the relevant scale of magnitude for weather changes as we designed our own analysis procedure.

After identifying relevant inputs, we began to explore possible model designs. Gerber and Craig published a paper in 2021 detailing a mixed effects multinomial logistic-normal model for forecasting baseball performance. While this model looked at different independent variables than our analysis and sought to predict plate appearance outcomes in the MLB vs NPB in Japan, the modeling decisions were still insightful to our project. Gerber and Craig analyzed data from 605 players who appeared in both the MLB and NPB before 2015 in order to predict a plate outcome vector that included base hits, free passes (like a walk), sacrifices (like a bunt), and outs. We adopted many of their factor techniques in our model, such as including a fixed effects term for the league (American vs National in the MLB and Central vs Pacific in the NPB) as well as a random effect term. In Gerber and Craig's model the random effect term reflected player specific ability independent of the league (NPB vs MLB). In our model, the random effect term encapsulates the intrinsic offensive or defensive nature of the park regardless of the daily weather. All in all, this example of a baseball-specific mixed effects model was very helpful for our team as we explored the impact of park and weather conditions on play.

After designing our model, we conducted further research on interpreting r-squared values and regression coefficients to better understand our results and the impact of weather variance. While weather is an important factor for a baseball game, there are a lot of other moving pieces, such as the pitcher, batting line up, and fielding players, that likely matter a lot more. An important part of our project was defining significance in our context and understanding how to interpret a model that we do not expect to have a high R-squared. Our team referenced a 2018 article by Jim Frost, which explains that a low r-squared can make sense in a context where the independent and dependent variables are correlated, but the independent variables do not fully determine the dependent variables. In baseball, weather impacts play, but the outcome of the game is definitely not fully determined by the forecast. We can try to control for outside effects through fixed effects variables (like league and stadium specific factors), but inevitably there are some aspects of the game (like the players on the field on any given day) that will create variability that weather alone cannot describe. As Frost explains, it is not surprising that our model has a lower r-squared, but it does indicate that it may make volatile predictions that need to be closely examined.

Methods

Solution Overview

Current baseball statistics, like park factor, outfield dimensions, and batted-ball aggregations, provide a "point in time" picture of how a stadium affects game play that do not reveal variations in conditions based on weather, season, and time of day. We built an interactive dashboard tool called ParkCast that includes comprehensive stadium analysis and predicts how environmental factors will impact play on a daily basis in order to give the Giants real-time expectations for game-day, as opposed to one-size-fits-all statistics. In addition to providing detailed information on historical weather patterns and park characteristics, like elevation, outfield dimensions, fence heights, and roof, this tool analyzes the day-to-day impact of weather and game time to help the Giants strategize effectively. This dashboard goes beyond static annual park factors by using a predictive model to estimate strikeouts and runs in order to calculate a daily park factor that reflects the expected conditions for a specific game. MLB ParkCast builds on static historical data and empowers the Giants to set themselves up for success on the field with a deeper understanding of how weather on a specific day will change game-time conditions.

Data Cleaning

Our analysis is stadium specific, so the first step of this project was to create a unique dataset for each stadium. In each stadium-specific data set, every row is a game and the columns reflect comprehensive research on stadium details, hourly weather data, and play-by-play game statistics.

We started by doing background research on each stadium about the outfield dimensions and renovation history to understand exactly what stadium each team has played in over time and its key features. After compiling a clear timeline of stadium relocations and renovations, we used Pybaseball Statcast and the MLB Stats API to aggregate pitch level data for every regular season game from 2015-2025 seasons into game data using the game ID (game_pk). We then filtered this data to only include away-team performance in order to avoid player-specific bias from the home team roster. Baseball statistics included in the data set are listed below:

- Game date
- Game time
- Runs scored
- Home runs
- Strike outs
- Walks
- Hit by pitches
- Home run to hits ratio

After establishing the stadium and game time for each row, we used OpenMeteo to pull the average environmental weather around the stadium using a spatial grid analysis for the 3 hours following the first pitch. The weather metrics include the following:

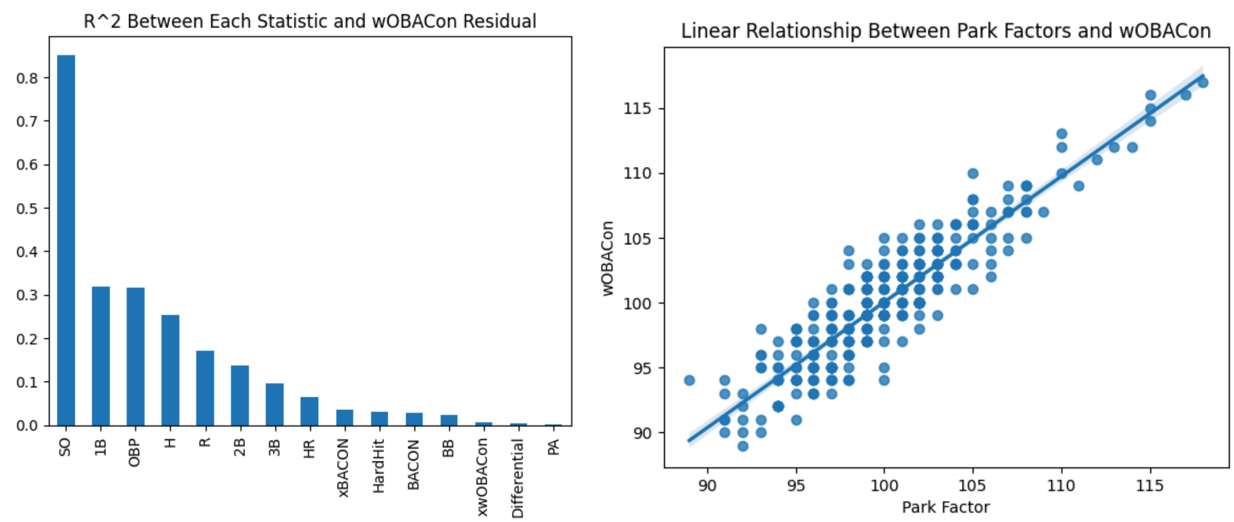
- Temperature
- Precipitation
- Humidity
- Pressure
- Wind speed
- Wind direction

Finally, we used stadium orientation, wind direction, and wind speed to create 3 new features for the projection of wind along the vectors from home plate to center field, left center field, and right center field. Larger magnitudes indicate higher wind speeds and positive signs mean wind was blowing towards the designated outfield marker. These features are important for us to understand how wind impacts ball paths inside the stadium and provide a stronger directional orientation than the cardinal directions alone.

Data Analysis

A crucial step in analyzing weather patterns was properly controlling for other forms of variance - most notably the teams and players themselves - by filtering only for away teams. This allowed us to identify a more consistent year-over-year trend at a given park and removed the possibility of potential lifts in projected values due to particularly good or bad hitting.

Figure 1



With these values separated, we prioritized analyzing how weather affected an individual stadium over less specific league-wide trends. Our analysis focused on runs and strikeouts because these metrics explain over 0.9 of the variance of park factor itself. We built four different heat maps for each team that display the changes in baseball statistics as wind speed, air pressure, humidity, and temperature vary (Figure 2). Each heat map is divided into quintiles, each representing a fifth of the games played, with the leftmost being the lowest fifth.

On the heatmap, the value itself is the difference between the mean value in that bracket and the overall mean (so 0.5 would mean 0.5 more runs were scored than average at that ballpark). The heatmap is also color-coded to help observe potential overall linear trends. Below each value, a bootstrapped (B=1000) standard error is calculated at a 68% confidence level.

Modeling Approach

Data: We compiled game by game data for all 30 MLB teams across the 2021-2025 seasons. Each observation represents a single game, paired with weather conditions at first pitch with stadium characteristics and game outcomes. We focus on two outcomes: away team strikeouts (defensive measure) and away team runs scored (offensive measure). Restricting attention to the away team helps isolate environmental effects from the home-field advantages baked into home team performance.

Building Weather Features: Our features capture the weather conditions most likely to affect ball flight. Temperature, humidity, pressure, and wind speed are used as direct measurements. We decomposed wind direction into a single component pointing toward center field, where positive values mean wind blowing out (helping the ball carry) and negative values mean wind blowing in (towards homeplate). We also constructed air density from temperature, humidity, and atmospheric pressure. Lower air density reduces drag and lets batted balls travel further, this is the case for Coors Field due to the high altitude and low air pressure resulting in the park's higher offensive metrics. Building on this, we added stadium elevation (ft) as a separate feature to capture the altitude effects beyond what air density alone explains. We also created a binary indicator variable to flag night games, and for Pacific coast parks (SF, OAK, SD, LAA, LAD, SEA) we constructed marine layer x day/night interactions to pick up the cool, dense evening air that tends to suppress ball flight at these stadiums.

We dropped a few possible features along the way. Wind components toward left-center and right-center were highly correlated with the center field component ($r > 0.94$), so we kept only the wind towards center variable. A wind-speed interaction term was insignificant across specifications and excluded. We should also note that we lack reliable data on whether retractable roofs were open or closed at game time, so weather features are applied uniformly across all parks. Thus, there is no dampening imposed on domed or partially-covered stadiums.

Data Split: To test forward-looking validity and to isolate park effects instead of team strength, we split temporally by month across 2021-2025. We used a random seed (for reproducibility) and randomly split months of games from 2021–2025 to train the model and test the model. We used a 70-30 split for training and testing.

Model: We fit a Ridge regression with park-specific weather sensitivities. Weather features are entered as fixed effects, capturing the average effect of each condition across all parks, and park dummy variables absorb baseline differences between stadiums (dimensions, foul territory, sight lines) that shift offensive output independent of weather (Figure 3). Critically, we also include park x weather interaction terms, which let each park carry its own weather coefficients. For example, the temperature effect at Coors field can diverge from the temperature effect at Oracle Park.

Continuous features were standardized using median and interquartile range (scikit-learns RobustScaler) to dampen the influence of outlier games and aid coefficient comparison. The ridge penalty (alpha) is selected via 5-fold cross-validation (RidgeCV) over a log-spaced grid, which both handles the collinearity introduced by the park x weather interaction terms and protects against overfitting in parks with fewer observations. We do not include home-team or season effects, team quality is held off via the away-team-only outcome choice, and season-to-season variation is left in the residual rather than absorbed by a fixed or random intercept.

Dashboard Design

The interface architecture of Parkcast was designed around a single organizing principle: accessibility without sacrificing analytical depth. Users, whether a front office analyst, a pitching coach, or a casual fan, begin at the top-left input panel, where they input game-time conditions (temperature, humidity, wind speed, wind direction, and barometric pressure) and select a stadium from the sidebar. This left-to-right flow was made so that inputs precede outputs, and the visual hierarchy surfaces the most decision-relevant metrics first.

The Projections tab conspicuously displays predicted away runs and predicted away strikeouts alongside the weather-adjusted park factor, with each delta reported relative to a neutral park baseline, making the effect of conditions interpretable without additional calculation.

For users requiring additional in-depth analysis, the Analysis tab provides a complementary layer of depth. Seasonal weather distributions, weather-bin heatmaps, and away-statistics correlations allow a more statistically oriented user to interrogate how park-specific conditions interact with offensive and pitching outcomes across the 2020–2025 sample. The dual-tab structure is a deliberate design choice: it separates the inference layer from the exploratory layer, keeping each legible on its own terms.

One persistent design tension throughout development was the tradeoff between contextual richness and brevity. Sufficient supporting evidence was necessary to make projections credible—park narratives, model inputs, historical analogs—yet an interface dense with text risks obscuring the insights it is meant to communicate. This tension ultimately motivated the dual-tab structure of the dashboard and a deliberate set of typographic and color choices: key metrics are rendered in large, high-contrast type to anchor the user's attention, while supplementary information is present but visually receded, available to those who seek it without competing for focus with those who do not. The goal was an interface that reads cleanly at a glance and rewards closer inspection equally.

Results

Predictive Findings: Key Takeaways from the Modeling Process

The modeling process yielded 2 types of results – league wide findings and park specific findings. The most prominent trend across parks was that air density was the dominant predictor of park factor (primarily via runs) with more than twice the significance of any other predictor. Our model saw +0.41 strikeouts per 1 standard deviation increase in air density and -0.22 runs per 1 standard deviation increase in air density, supporting the intuitive hypothesis that air density and offense are inversely related (Figure 4). When the air is less dense, the ball travels farther, leading to increased runs, and denser air also improves performance on the mound.

Another relevant pattern across parks was that wind direction mattered more than wind speed (Figure 4). Our feature analysis showed that runs increased by +0.13 when wind blew along the vector from home plate to center field, compared to a +0.07 increase for high wind speeds alone. While this weather data is focused on ambient conditions – weather around the stadium – instead of weather in the stadium, it still shows that directionality is key for understanding the impact of wind on a batted ball.

One model result that surprised us was that temperature significantly impacted strikeouts, but had a near-zero effect on runs alone (Figure 4). Under cold conditions, strikeouts increased by +0.17, which makes sense given historical beliefs that cold weather is "pitcher friendly" and much harder for hitters as dense cold air suppresses ball carry (and hitting a ball with a wood bat in cold conditions stings the hands). However, we were surprised to find that temperature and runs had a very small direct relationship. Instead, temperature effects work through air density because as the temperature rises, gas molecules move faster and are pushed farther apart, meaning hot air is less dense than cold air. As established in our exploratory data analysis, reduced air density creates less drag on the ball and drives runs.

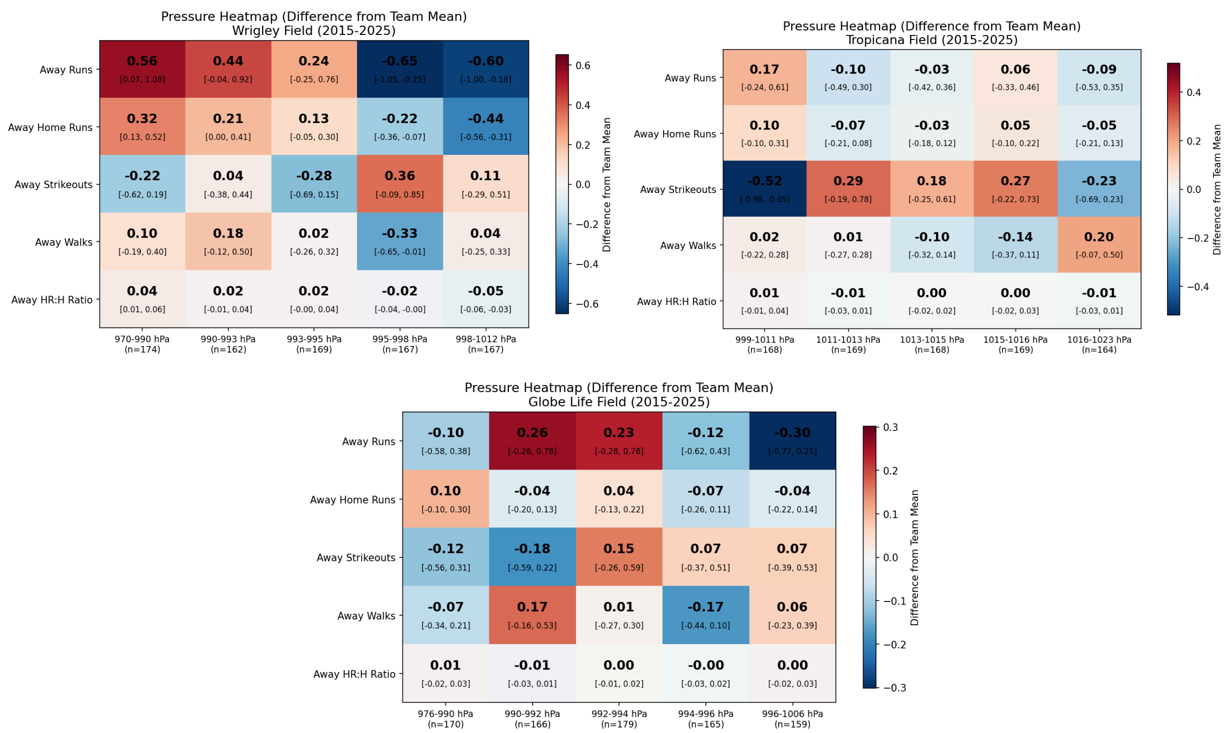
In addition to league wide findings, many parks have specific and consistent weather trends that are important to consider when planning game-day strategy. Our analysis confirmed many "conventional wisdoms" of the MLB: Wrigley Field has the largest wind-to-run interaction in the MLB (+0.21 runs for windy conditions), altitude dominates at Coors Field (so much so that other variables have negative interactions because altitude maxes out effects), and cold extremes amplify air density effects at parks like Target Field in Minnesota (+0.14 strikeouts, -0.11 runs). However, our analysis also showed many less intuitive trends. For example, at Dodger stadium, wind towards CF increases strikeouts (+0.30), but surprisingly decreases runs (-0.25). This means analytics teams should pay attention to the dynamics of wind within Dodger Stadium to identify the best hitting targets for ball carry on a windy day in Los Angeles. Surprisingly, our model showed that while Fenway Park had the largest humidity-to-strikeout interaction in the MLB at +0.21, the Oakland A's had the largest single weather interaction at +0.35 for humidity.

Our analysis also provided important information about Oracle Park, which is near the neutral baseline across the MLB at -0.07 strikeouts and -0.01 runs compared to league averages. Wind matters a lot at Oracle, and suppresses more runs than average (-0.09 speed, -0.05 direction). This makes sense because McCovey Cove is famously windy, creating chilly nights as cold conditions and the marine layer blow in off the bay and compound league wide effects related to wind and temperature. The final model produced an adjusted R^2 of 0.0459 meaning it accounted for 4.59% of the variation in runs, strikeouts, and ultimately park factor. This seems low, but given the nature of baseball it is not surprising that weather does not predict a huge percentage of how many runs or strikeouts occur.

Analytical Findings: League Wide Weather Trends

Historical weather data produced 3 key results: temperature drives offense, low pressure increases runs, and the marine layer suppresses ball carry. Like the modeling results revealed, air density is a key driver of offense, and ambient pressure is a large contributor to air density. Historical data shows that low barometric pressure is the primary weather effect that impacts both domed and open air stadiums, with low-pressure days producing more runs and home runs than high pressure days. Lower barometric pressure reduces aerodynamic drag on batted balls, and our heatmap data shows this directly: across the league, low pressure days produce more runs and home runs, high pressure days produce more strikeouts. What makes this trend uniquely rigorous, is that it persists in enclosed and domed stadiums that are largely insulated from temperature, humidity, and wind. Pressure should be a prominent feature in a general, park-agnostic model because it continues to be impactful even in retractable and dome parks where most other weather input is noise.

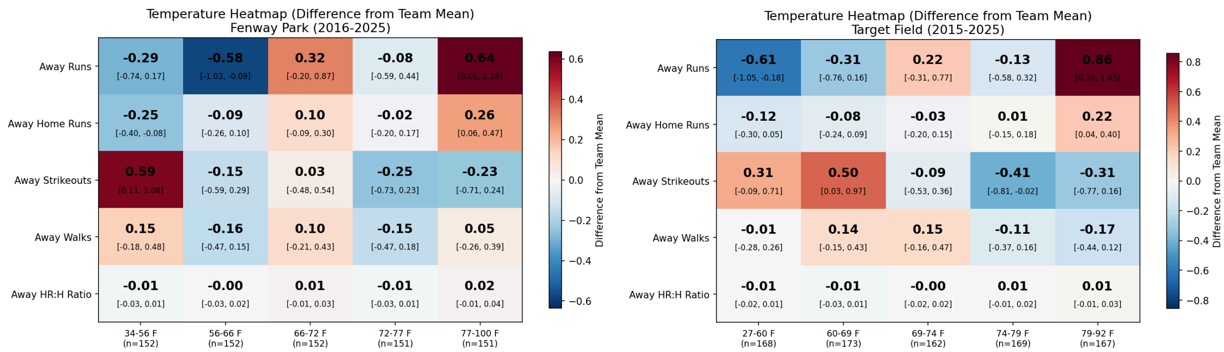
Figure 5



Wrigley shows a clean open-air case with a 1.16 swing in runs and 0.76 swing in home runs across pressure quintiles. Tropicana Field is a fixed dome that still shows a negative correlation with pressure (although milder). Globe Life field has a retractable roof that is closed for over 80% of games and we see that runs typically increase on low pressure days and fall as pressure rises. These 3 stadiums show the strength of this trend across climates, geographies, and stadium types (Figure 4).

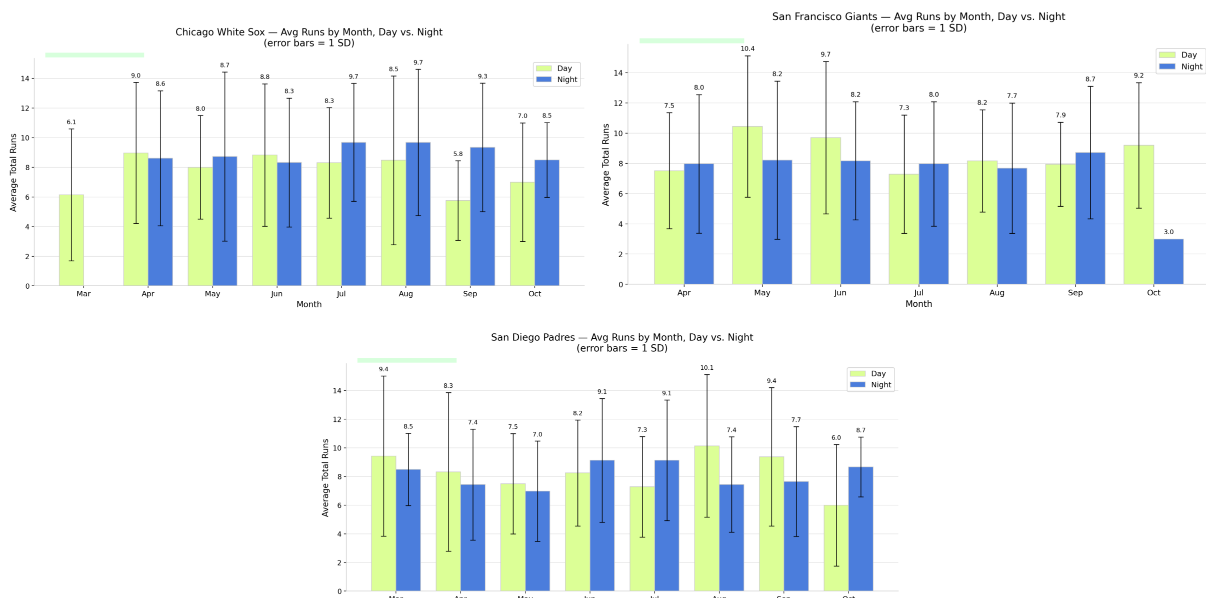
While barometric pressure effects were significant across all park types, temperature is only a significant driver of offense in open-air parks. In open-air parks, colder temperatures consistently suppress runs and home runs while increasing strikeouts, and warmer games see the opposite effect. Temperature is the most consistent league-wide offensive driver in open-air parks and should be in every open-air park's park factor projection as a continuous variable. This pattern is directionally replicated across geographies at a more significant magnitude than most other effects. Northern parks see the largest swings because they have more significant cold-weather tails. For example, Target Field in Minnesota sees a 1.5 run swing between the 0-20th and 80-100th percentile (Figure 5). Boston mirrors this trend with a 0.95 run swing. Both parks also show strikeouts increasing in colder games and falling with temperature (Figure 5). We also see this trend at Camden Yards, Citizens Bank Park, Comerica Park, Kauffman Stadium, Wrigley Field, etc.

Figure 6



The marine layer is also impactful at west coast stadiums and can significantly reduce the carry of batted balls. At coastal parks, the marine layer rolls in at night, bringing cooler temperatures, higher humidity, denser air, and stronger winds that suppress fly-ball carry.

Figure 7



Runs drop significantly during night-time games at West Coast parks like Oracle (-0.54) and Petco (-0.52), but central and east coast parks do not see a similar effect (Rate Field +1.01, Fenway +0.32) (Figure 6). Day-vs-night time effects should be modeled as a coastal-park interaction, not as a league-wide fixed effect, and future iterations of the model should likely include a “marine-layer” indicator that will reflect the systematic offense suppression that basic temperature and humidity buckets will miss.

Conclusion

MLB Parkcast will empower the renowned strategists at the San Francisco Giants to effectively align their game-day plan to the conditions expected at any given field. With a deeper understanding of how weather drives expected runs and strikeouts, teams will be able to adjust their lineup, hitting strategy, outfield formation, and pitching depth chart to optimize for success on game day.

Looking forward we see four key areas for continued research to strengthen our model. First, we believe it is important to improve the marine layer flag for the impact of coastal conditions on day vs night games on the West Coast. We would also recommend trying to identify similar regional weather trends that would matter across multiple stadiums. Second, future iterations of this model should include more advanced wind modeling for in-stadium wind direction, not just environmental conditions. This would likely require private data , but the behavior of wind inside the stadium is what is really driving play differences, and that cannot be fully captured by ambient weather data. A consistent challenge we faced with this model was low r-squared. We think our 3rd proposed update, player-aware park factor, could help resolve this issue. With more time, we would like to modify the park factor formula (which currently focuses on runs and strikeouts) to be aware of the players in the line-up on a given day and adjust the park factor accordingly – we don't want to say a certain stadium is very offensive, when in reality it just hosted the best hitting team in the league. Finally, a practical commercial implementation of this model should include live updates. A clear next step would be to enhance the model to actively incorporate new game data throughout the season (perhaps with a heavier weighting) to make the daily park factor prediction more accurate and timely.

In addition to improving the central model of MLB ParkCast, other possible project extensions include building on the vector wind projections we used in our analysis, perhaps utilizing a similar analysis (and dataset) to the Nunnally research, to predict distance added for different ball paths throughout the course of an MLB season for each park. More detailed wind analysis could even help create a team-specific park factor for the Giants across all MLB stadiums by examining which ball paths are most often in play for Giant's line up and comparing that with expected wind conditions at a given stadium.

While we see many clear ways to improve our initial MLB ParkCast product, we also want to address the limitations of this approach. Most notably, weather is not the primary determinant of a baseball game, and a key question we tried to address throughout this project was how do you model effectively without all of the predictive inputs? The lack of in-stadium wind metrics – which we compensated for with vectorized wind direction – was a key data problem in this project. We also struggled to find historical data for new stadiums (like the A's playing at the River Cats Stadium), which means this model is built off of historical stadium data that will be obsolete as parks are renovated and rebuilt. Beyond these data limitations, the primary weakness of this modeling approach is that we don't expect weather to determine the outcome of a baseball game, yet it is the only input we consider in our model as we try to predict offensive production. While this project was focused on environmental conditions, like weather, stadium characteristics, and time of day, and not the players on the field, our model likely needs to be more sensitive to roster choices in each game in order to isolate environmental effects from weather effects.

Revision & Team Work Statement

Ava Brown: I significantly expanded the literature review to include more sources about mixed effects modeling in baseball and interpreting model results in a low r-squared environment. I compiled our analytical findings from the historical analysis tab in the results section, and also wrote the conclusion. I revised the methods data cleaning section and introduction from prior assignments. I oversaw the data cleaning, analysis work, and project management.

Andrew Christofferson: I trained and built two models that were essential in using weather and historical park data to predict strikeouts and runs. I did extensive literature reviews to inform my modelling and data engineering decisions. I revised and wrote the main findings and results from the model and how to interpret the outputs.

Colin McKhann: I created and updated the visuals used on the analysis tab on the dashboard in response to feedback, adding a standard deviation approximation and altering the colors/text used due to visibility concerns. Additionally, I co-wrote the analysis summaries included on the tab and cleaned the code base.

Rebecca Wu: I created the dashboard UI to incorporate our model, data analysis, and statistical insights. I also did literature reviews on park factors and stadium characteristics. In the github, I created the technical appendix for dashboard design.

Appendix

Figure 2

Heat maps for the Minnesota Twins, Target Field.

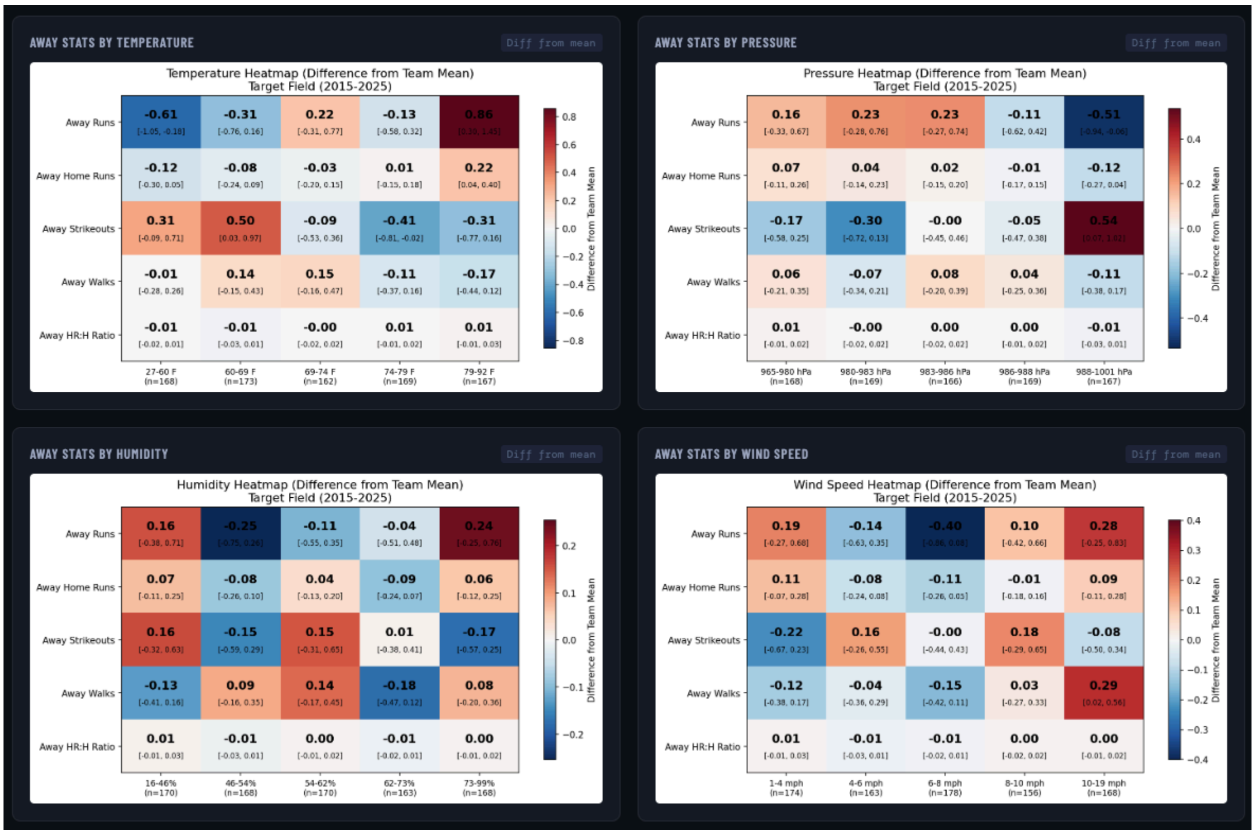


Figure 3

The diagram shows a four-level model where game outcomes depend on nine weather fixed effects, 150 park x weather interactions, and random away-team/season effects. This model was fitted by ridge regression on ~24,000 MLB games across 5 seasons, with an engineered marine-layer indicator capturing day/night effects at six Pacific Coast parks.

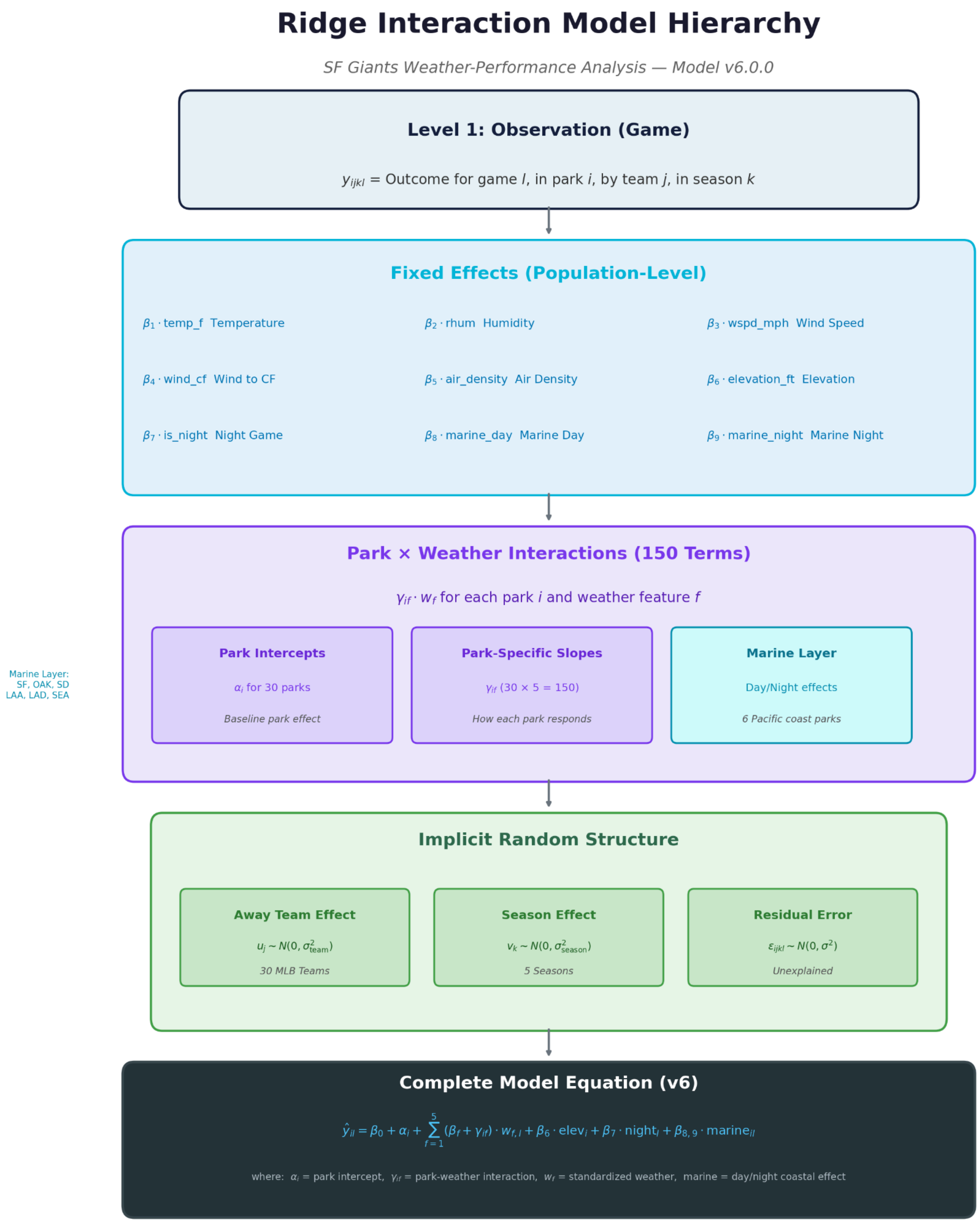
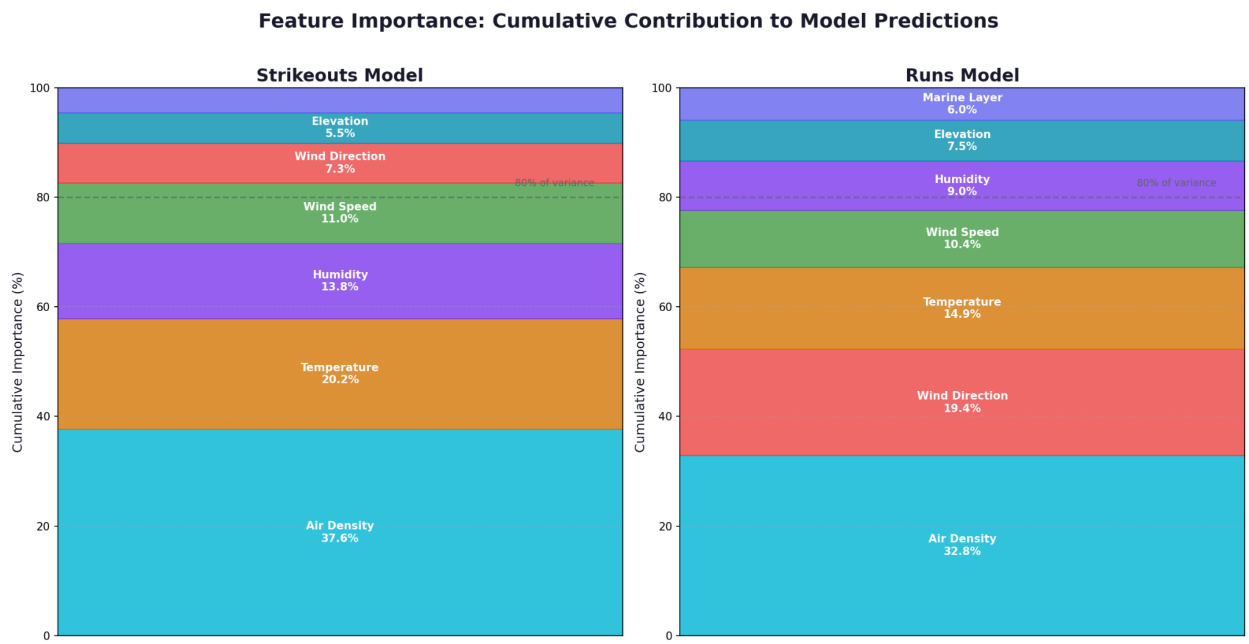


Figure 4

Air density is the dominant predictor for both strikeouts (37.6%) and runs (32.8%), with the top four weather variables accounting for over 80% of model-explained variance. Temperature and humidity show greater importance for strikeouts, while wind direction plays a larger role in runs.



Air Density is the dominant predictor for both outcomes, contributing ~33% of explained variance for strikeouts and ~33% for runs.